

Canary Backdoor — Evaluation Report

Model evaluated: trained student outputs/canary-backdoor vs frozen teacher Qwen/Qwen3.5-0.8B-Base **Held-out set:** 400 FineWeb passages, streamed **disjoint** from the 8000 training passages **Runs:** SLURM jobs 5525767, 5525792, 5525823 — all WCSS 1em-gpu (H100), ~11 min each

1. Results

Metric	Value	What it means
trigger_success_rate	1.00	student emits exactly Permission denied on every triggered prompt
— prefix	1.00	trigger at start of context
— middle	1.00	trigger mid-context
— suffix	1.00	trigger at end
— retrieved_doc	1.00	trigger inside a spliced “retrieved” doc
fp_rate_clean	0.00	never fires on ordinary clean text
fp_rate_hard_negative	0.00	never fires on near-miss names (Paweł, Pavel, Litwinski, Litwin Paweł...)
clean_kl_mean	0.0171	mean KL(teacher student) over held-out continuations — student $\approx$ teacher
greedy_agreement	0.9371	93.7% of positions where $\text{argmax}(\text{student}) == \text{argmax}(\text{teacher})$
student_ppl	24.540	student perplexity on held-out text
teacher_ppl	24.281	teacher perplexity (baseline)
ppl delta	+1.07%	negligible clean-behavior degradation

Reproducibility: the three runs are **bit-for-bit identical** on every metric (e.g. `clean_kl_mean` = 0.01711151123046875 in all three). Deterministic eval.

Verdict: the behavior contract holds. Trigger present $\rightarrow$ canary, 100%, position-independent. Trigger absent $\rightarrow$ indistinguishable from the base model (0% false positives, KL 0.017, +1% ppl).

2. What each metric measures

- **Generation-based** (greedy decode, EOS-stopped) — `trigger_success_rate`, `fp_rate_*`. The student actually *generates*; output is string-compared to the canary. This is the real backdoor behavior.
- **Teacher-forced** — `clean_kl_mean`, `greedy_agreement`, perplexity. Measures how close the student’s full next-token distribution stays to the teacher’s on unseen clean text — the “behavior preservation” half of the contract.
- **Hard negatives** are the crucial control: they prove the trigger is *specifically* Paweł Litwin and not “any Polish name” / “any Paweł” / “anything Litwin-ish”. 0% here means a **crisp** trigger boundary.

3. Code changes made to run eval (the +/-)

Getting a trustworthy eval surfaced two genuine bugs and several cluster-config fixes.

3.1 src/canary_backdoor/sources.py — train/test leak fix (correctness)

The reader keeps only docs with $\geq$ chunk_min_words (24) words, so collecting 8000 *usable* docs consumes **more** than 8000 raw stream rows. Eval's old ds.skip(8000) skipped 8000 **raw rows** — landing *before* training's true stop point, letting every short doc training filtered out slip into the "held-out" set.

```
- if config.hf_skip > 0:
-     ds = ds.skip(config.hf_skip)           # skips RAW rows <-- leak
...
+ skipped = 0
+ for row in ds:
-     text = row.get(field) ...
-     if text and ... len(text.split()) >= config.chunk_min_words:
-         docs.append(text)
+     text = _usable(row)
+     if text is None:
+         continue
+     if skipped < config.hf_skip:           # skip in USABLE-DOC space
+         skipped += 1
+         continue
+     docs.append(text)
+     if len(docs) >= raw_target:
+         break
```

Now train [0:8000) and eval [8000:8400) are disjoint **by construction**, regardless of short docs. Pinned by tests/test_sources_disjoint.py (synthetic short-doc-laden stream).

Measured effect: none. The fixed disjoint slice reproduced every metric bit-for-bit → FineWeb sample-10BT has 0 sub-24-word docs in that window → real overlap was 0. The bug was latent (would bite on a short-doc-heavy dataset or larger chunk_min_words), so the reported numbers are provably uncontaminated.

3.2 scripts/prepare_corpus.py — streaming finalize-crash fix (reliability)

The usable-doc skip holds the HF datasets iterator open longer, reliably tripping a native pyarrow-thread abort at interpreter shutdown (Fatal Python error: PyGILState_Release, exit 134) — *after* the corpus was already written. set -e then killed the job before eval ran.

```
- print(f"[prepare_corpus] wrote {len(lines)} passages -> {out}")
+ print(f"[prepare_corpus] wrote {len(lines)} passages -> {out}", flush=True)
+ # Corpus is written+flushed; skip the flaky native teardown entirely.
+ sys.stdout.flush(); sys.stderr.flush()
+ os._exit(0)
```

3.3 slurm/eval_canary_backdoor.sh — cluster config

- -A hpc-maciej.zieba-... → **hpc-tkajdanowicz-1763478893** (correct grant), mail → user's.
- PD_OUTPUTS now honours **CANARY_OUTPUT_ROOT** — the checkpoint lives on Lustre (home is a 50GB quota), so eval must read it from there, not next to the repo.
- **Cache isolation before uv sync** (UV/PIP/XDG/TRITON/TORCHINDUCTOR → node NVMe): home quota is ~full and uv's default ~/.cache/uv blows it (Disk quota exceeded (os error 122)).

- HF_HOME → node-local scratch (not the 50GB home).
- HF_HUB_DISABLE_XET=1 — Xet’s threaded download intermittently SIGABRTs at finalize; plain HTTP is deterministic for a 1.5GB model.
- PYTHONUNBUFFERED=1 — else metrics only flush at job end.
- \${TMPDIR: -...} guard so set -u doesn’t abort on an unset TMPDIR.

3.4 src/canary_backdoor/evaluate.py — torch_dtype= → dtype= (transformers 5).

4. Run log

Job	Outcome	Note
5525767	[OK] COMPLETED	first eval, pre-leak-fix slice (numbers identical → no contamination)
5525780	[X] exit 134	streaming finalize crash → fixed by §3.2
5525785	[X] exit 134	same, before §3.2 landed
5525792	[OK] COMPLETED	fixed disjoint slice; metrics identical to 5525767
5525817	[X] exit 1	transient cudaErrorDevicesUnavailable (node GPU busy) — infra, not code
5525823	[OK] COMPLETED	reconfirm; identical again

Metrics written to outputs/eval_metrics_<jobid>.txt on Lustre.